INF10025 Data Management and Analytics

Tutorial 7 – Discussion Questions for Week 7

Q1. How many rows are returned by this query?
SELECT COUNT(*)
FROM MOVIE

Q2. How many rows are returned by this query if there are NO rows in the MOVIE table?
SELECT COUNT(*)
FROM MOVIE

Q3. How many rows would you expect are returned by this query?
SELECT GENDER, COUNT(*)
FROM ACTOR

GROUP BY GENDER

Q4. What is wrong with this statement?
SELECT BIRTHCOUNTRY, COUNT(*)
FROM ACTOR

Q5. This statement will execute perfectly. Why is this statement next to useless?
SELECT COUNT(*)
FROM ACTOR

GROUP BY BIRTHCOUNTRY

Q6. What is wrong with this statement?

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SELECT BIRTHCOUNTRY, COUNT(*)
FROM ACTOR
GROUP BY GENDER
```

Q7. Provide a couple of example rows for the result set produced by this statement.
SELECT BIRTHCOUNTRY, GENDER, COUNT(*)
FROM ACTOR

GROUP BY BIRTHCOUNTRY, GENDER

Q8. Describe the purpose of the MIN aggregate function?

Q9. How many rows in the result for this statement?
SELECT MIN(BirthDate) FROM ACTOR

Consider this table named STUDENT

StuId	Name	Gender	BirthYear	Degree	SubjectsPassed
1	Fred	M	1991	BIT	0
2	Sue	F	1993	ICT	4
3	Emma	F	1992	BIS	2
4	Dave	M	1993	BIT	4
5	Leah	F	1991	ICT	6
6	Linda	F	1991	BIS	6
7	Harry	M	1992	BIS	4

Q10. Write a single SQL statement that counts the total number of students

Q11. Write a single SQL statement that counts the total number of students that are female

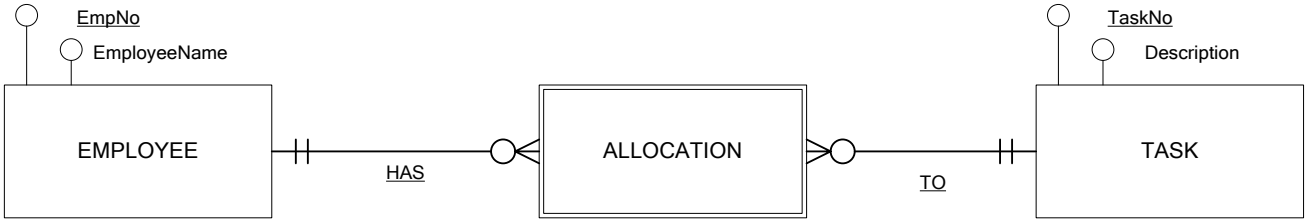
Q12. Write a **single** SQL statement that counts the total number of students of each gender
Hint: Use a group by clause

Q13. Write a **single** SQL statement that counts the total number of students of each degree
Hint: Use a group by clause

Q14. Write a **single** SQL statement that counts the total number of students of each degree.
Does not list any total values less than 2
Hint: Use a Having clause

Q15. Write a **single** SQL statement that calculates the average number of subjects passed by each Degree.
Only include Female students (ignore all males)
Does not list any total values less than 3
Hint: Use a Where, Group By and Having clause

Q16. Consider the following ERD



a. What is the identifier of the Allocation entity?

b. How many foreign keys will exist in the Allocation table?

c. Write the Relational Schema. Indicate all PKs & FKs.

d. Write SQL statements to

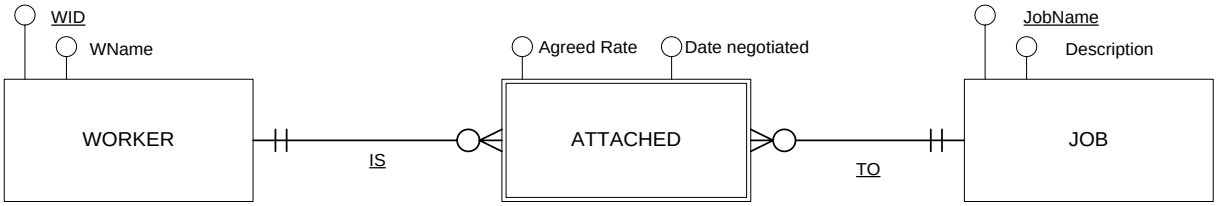
i. create employees Jim Black and Perry White

ii. create tasks Bake Cake and Chop Onions

iii. allocate Jim Black to the Chop Onions task

iv. allocate Perry White to the Bake Cake task

Q17. Consider the following ERD

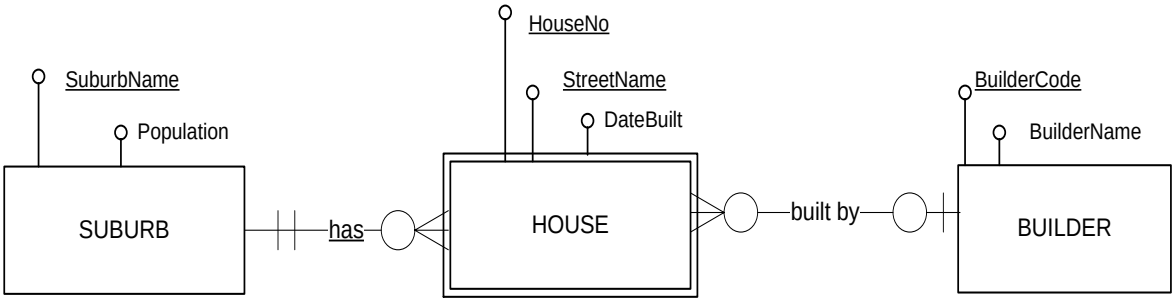


a. What is the identifier of the ATTACHED entity?

b. How many foreign keys will exist in the Allocation table?

c. Write the Relational Schema. Indicate all PKs & FKs.

Q18. Consider the following business case and diagram:



a. What is the Identifier of each entity
b. Convert the above ERD to a relational schema.